Reinventing Relevance Feedback in Search: Scalable, Cheap, Universal

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What's in this talk for each of you?

If you're a search engineer

- An overview of domain-specific research & a new method that closes a gap in the relevance feedback taxonomy

If you use retrieval in your pipelines (e.g. RAG)

- Approaches for improving the relevance of your retrieved results

If you're just curious

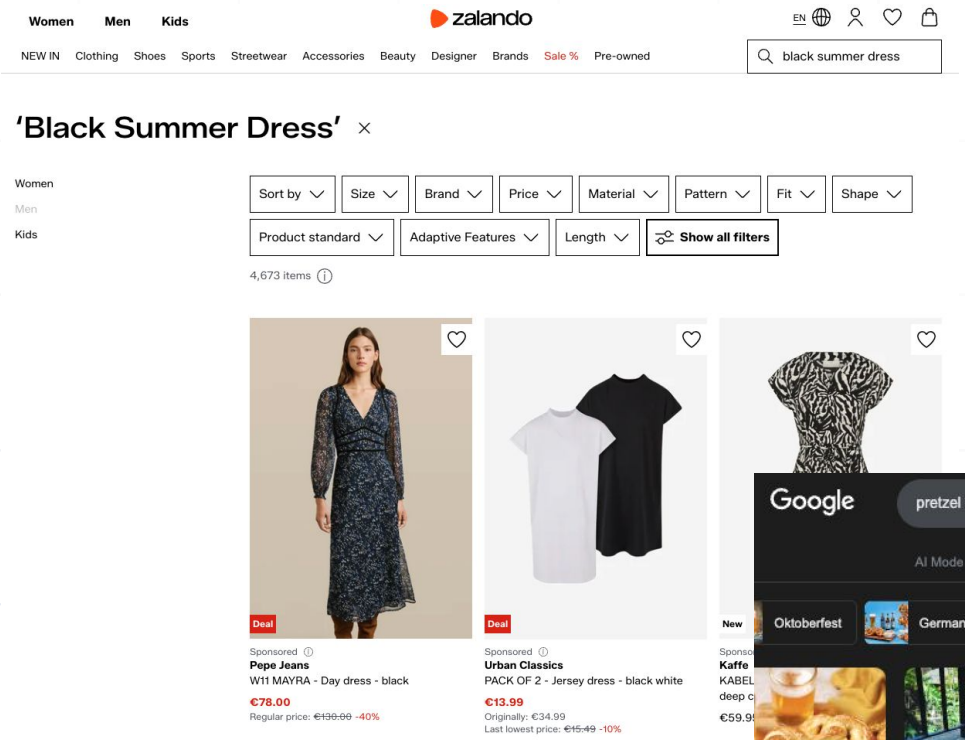
- The story of what happens when someone from industry goes looking for answers in academia

How we'll approach it

- **High-level overview (still a lot to introduce) —** all tech details in supplementary materials on slides
- **I like QR codes,** so be ready :)
- **All slides on GitHub after the meetup**, so don't worry if you skipped something
- **No questions are stupid questions!** Ask away or let's discuss at networking time:)

Search Relevance

Search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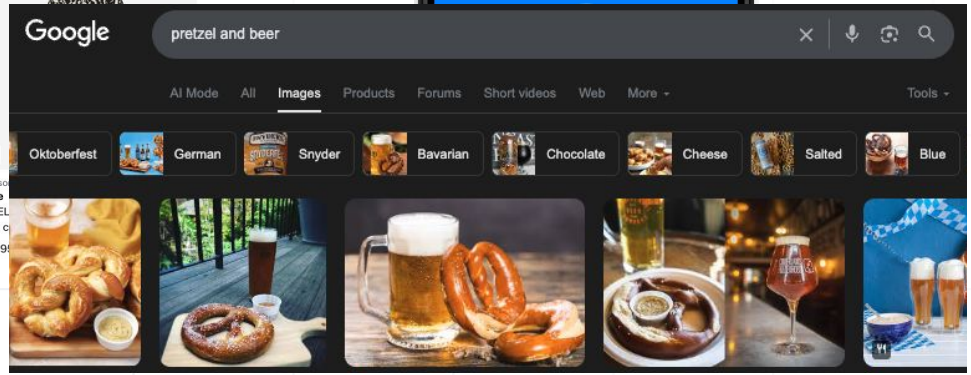
Length
Show all filters
4,673 items
Deal
Sponsored
Pepe Jeans
W11 MAYRA - Day dress - black
€78.00
Regular price: €130.00 -40%
Deal
Sponsored
Urban Classics
PACK OF 2 - Jersey dress - black white
€13.99
Originally: €34.99
Last lowest price: €15.49 -10%
New

9:30
My Shazam
Concerts
Tap to Shazam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Images
Products
Forums
Short videos
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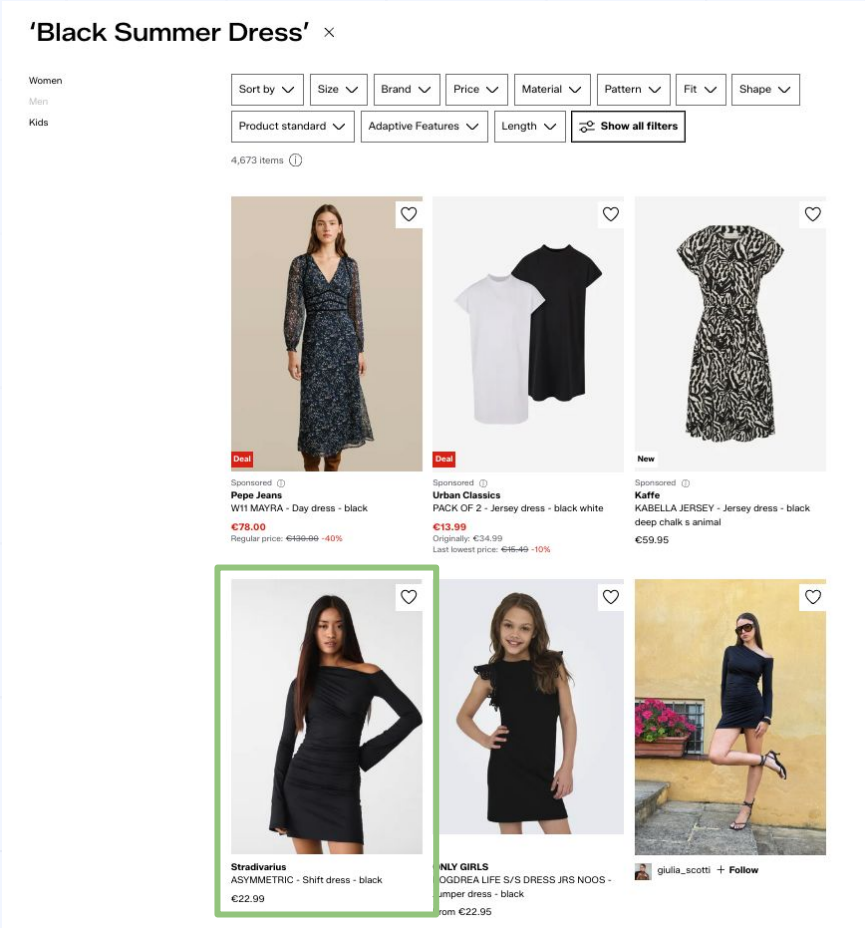
Oktoberfest
German
Snyder
Bavarian
Chocolate
Cheese
Salted
Blue

Search Relevance

How fitting the results are to the request, from the perspective of the search user

🔍 *The best meetup in the world?*



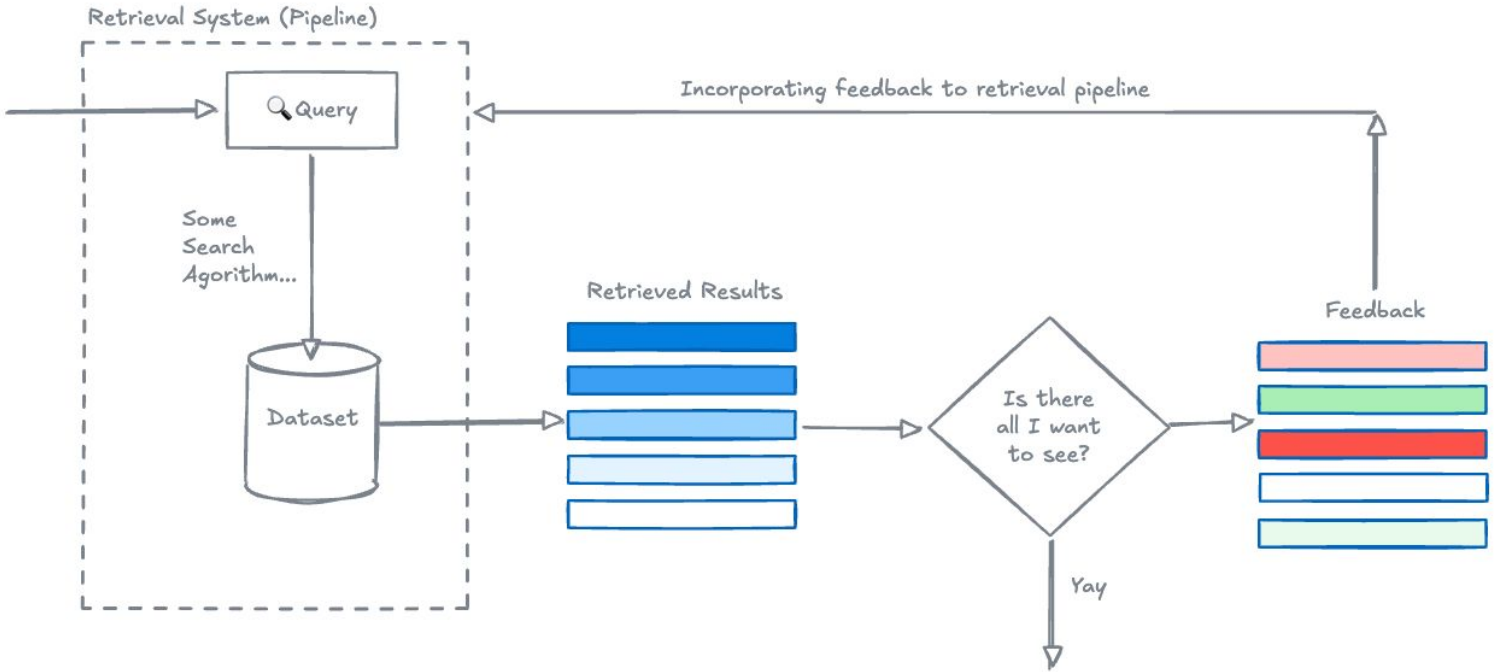
Search Can Return Irrelevant Results

- The **search algorithm** itself is not that good;
- The algorithm is generally ok, but doesn't reflect the user's understanding of relevance;
- The **search query** is not that good;
- The query is good, but not for this **dataset**: data is specific, skewed, etc...

...or just cause a bad day :(

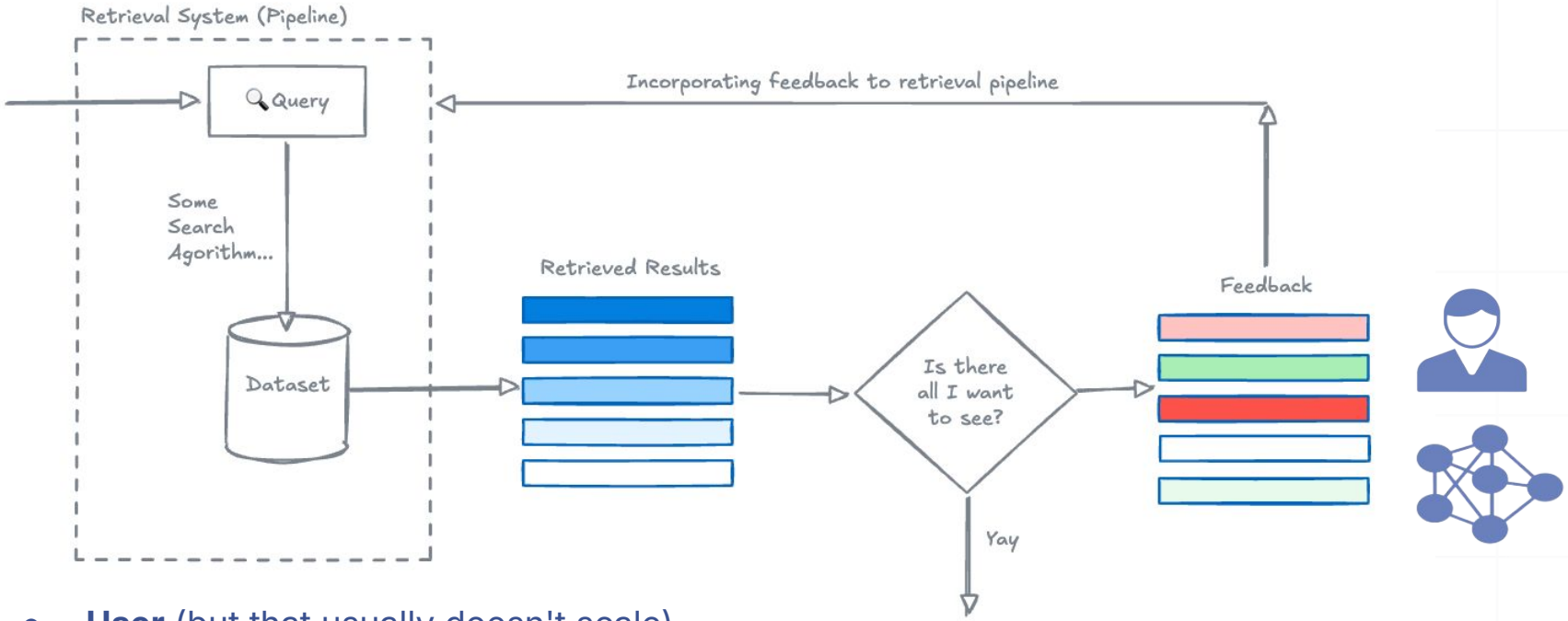
Improving Search Relevance: Relevance Feedback Approach

Feedback Loops in Search



Relevance Feedback – A mechanism for a retrieval system to **iteratively refine results in the direction of relevance**.

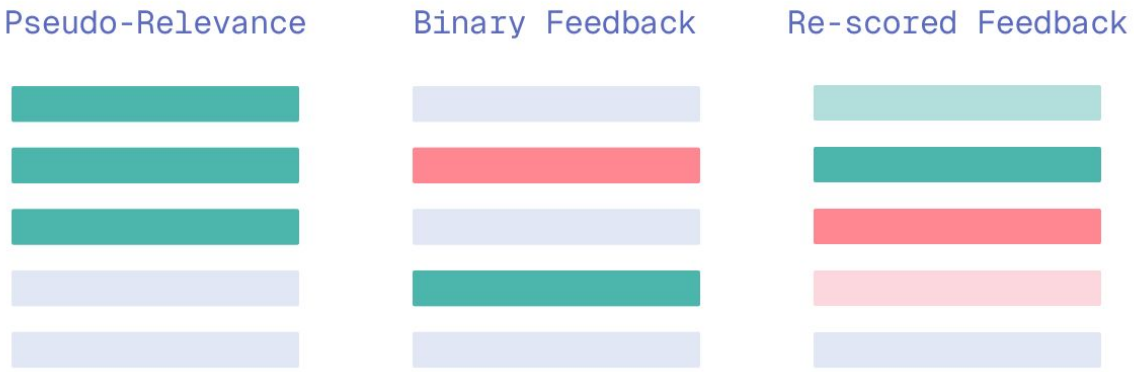
Who Provides Feedback?



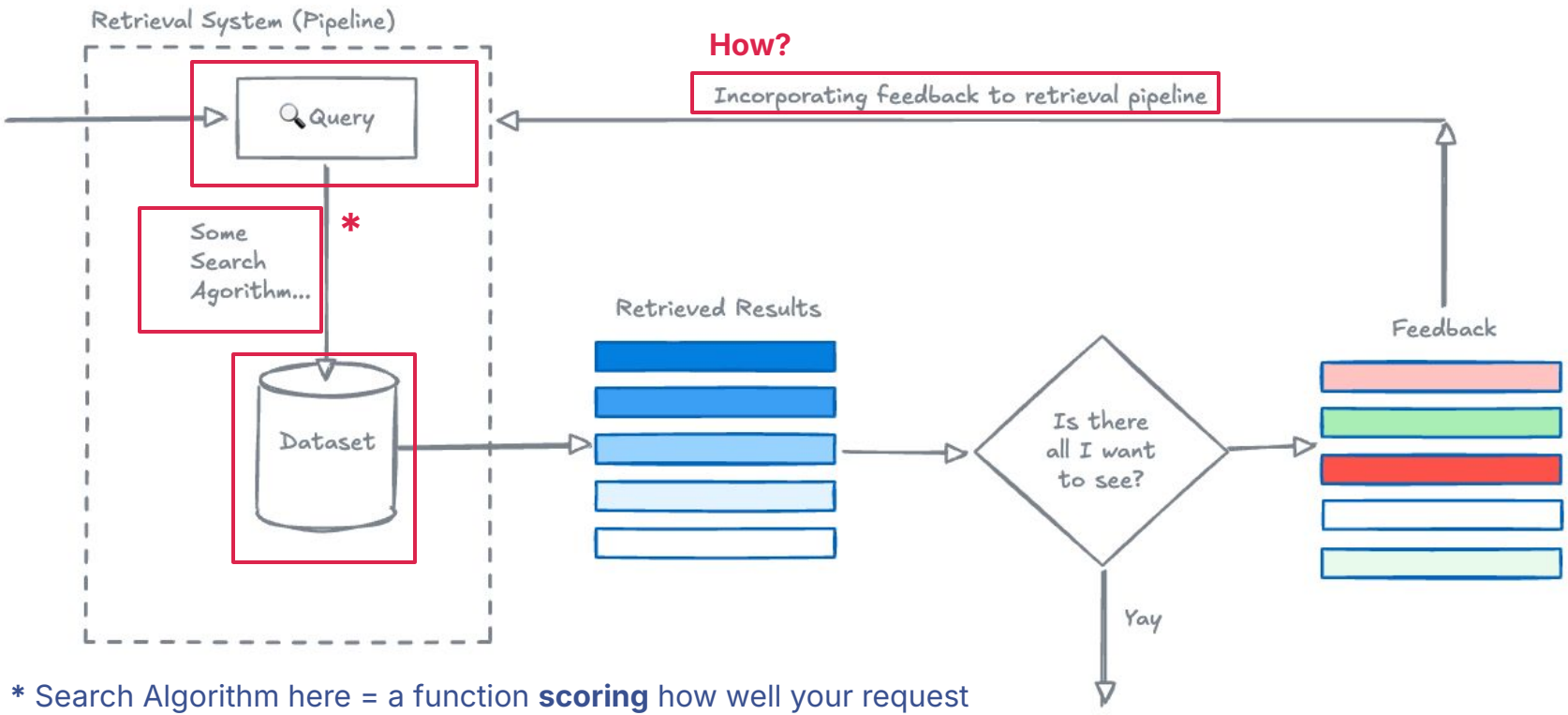
- **User** (but that usually doesn't scale)
- In production-level systems, usually a "**machine**": LLM, ML algorithm, reranker, heuristic/statistical function...

Which Form of Feedback?

- **Pseudo-Relevance Feedback** — assume top retrieved results are already relevant (*too basic:/*)
- **Binary** — relevant / not relevant (*too limiting:/*)
- **Graded (re-scored)** — a relevance score (higher = more relevant, *usually*) (*the best!*)

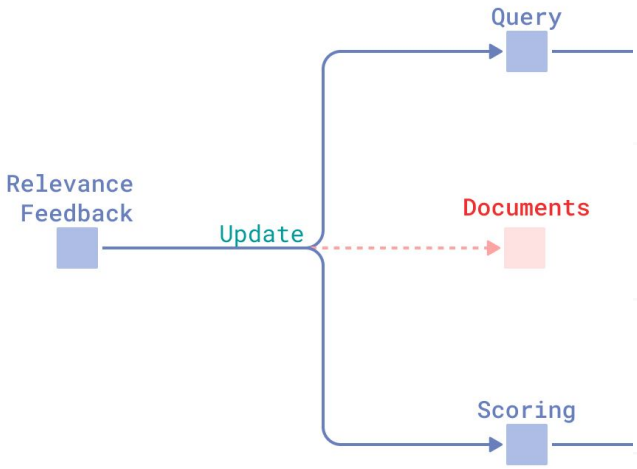


What to Do with Feedback?

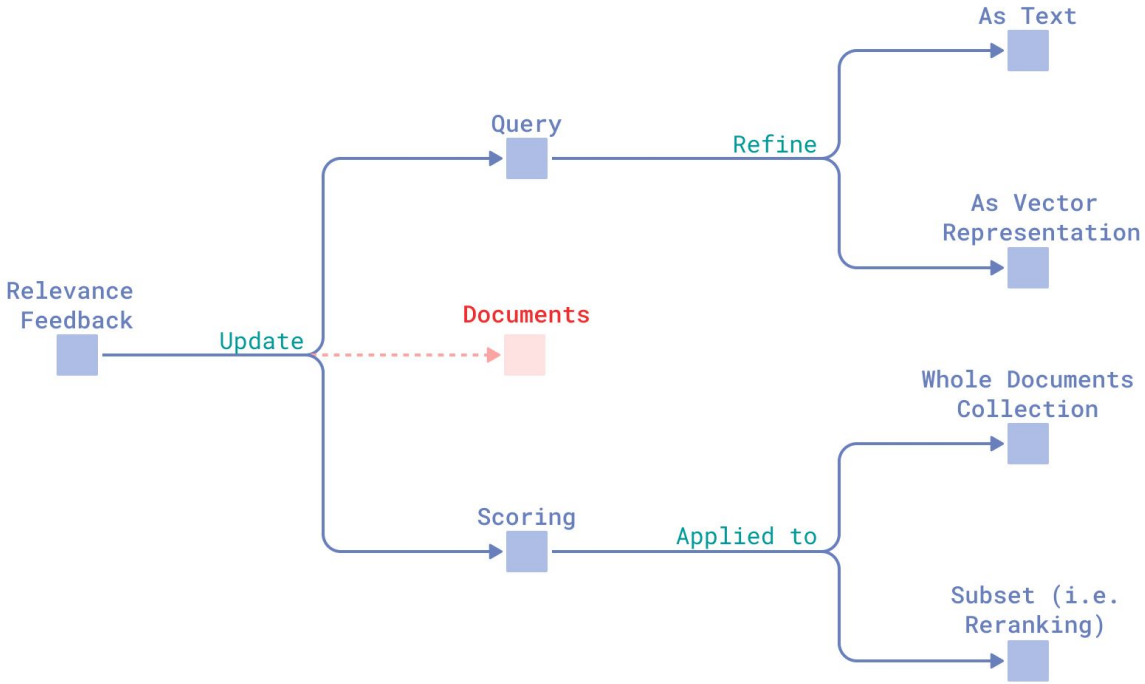


* Search Algorithm here = a function **scoring** how well your request matches your results

What to Do with Feedback?

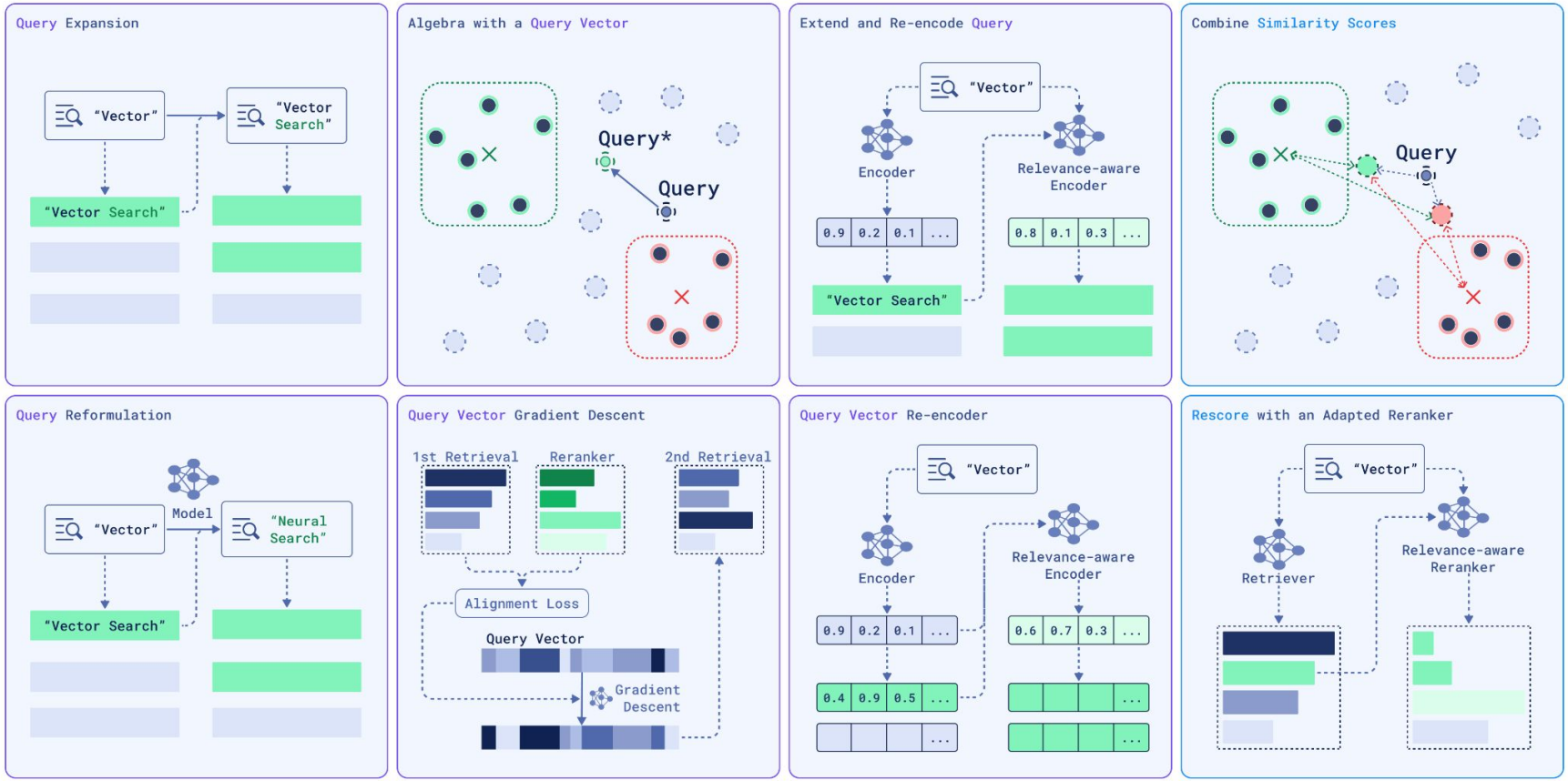


What to Do with Feedback?



Detailed Article on **Relevance Feedback Methods**

Visual Overview (*for the curious ones*)



My Goal: Relevance Feedback for Neural Search

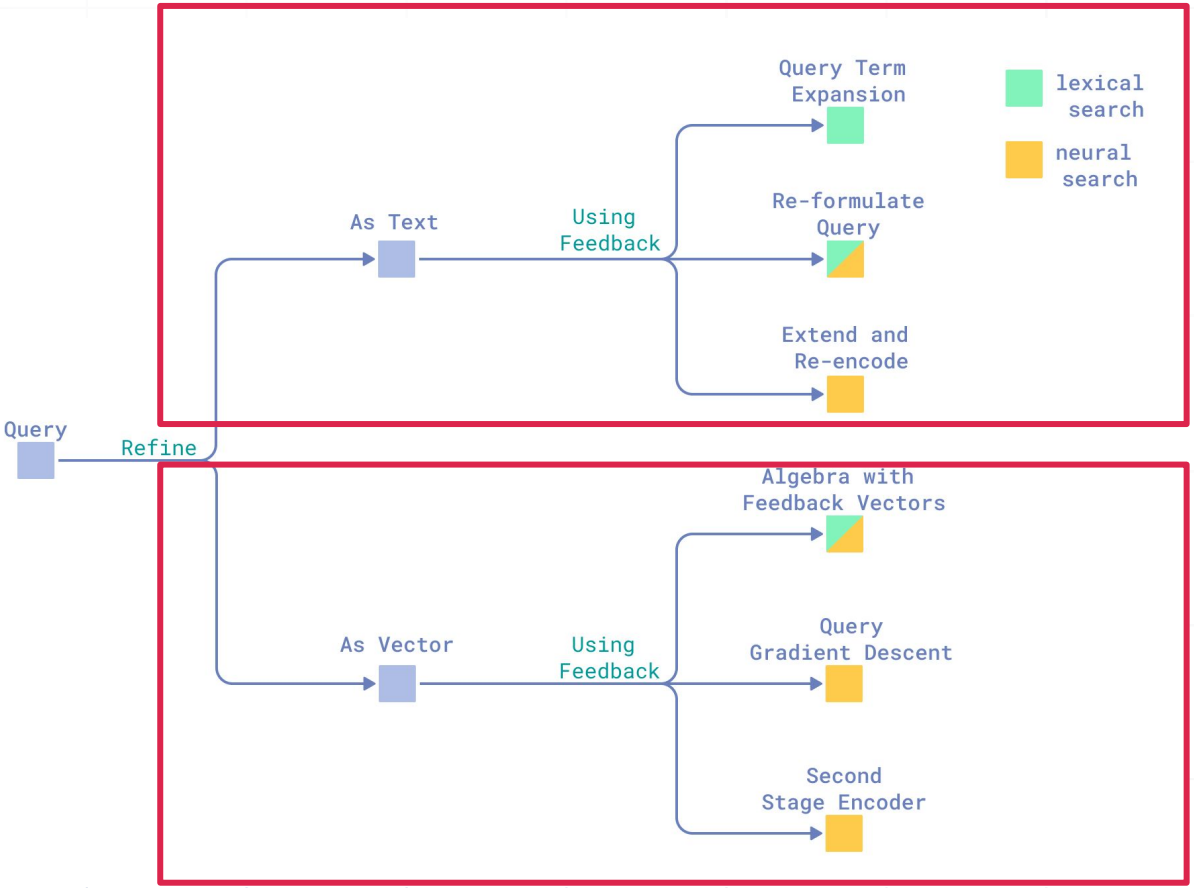
Neural search — works on **vectors** produced by ML models (embeddings),
Search is based on **semantic similarity** (usually measured by cosine distance).

Powers search across different data types (text, images, videos, diff languages, molecules...)

Lexical search — works on **symbols** (text),
Search is usually based on **statistical/probabilistic similarity** (how much do keywords in documents and queries overlap).

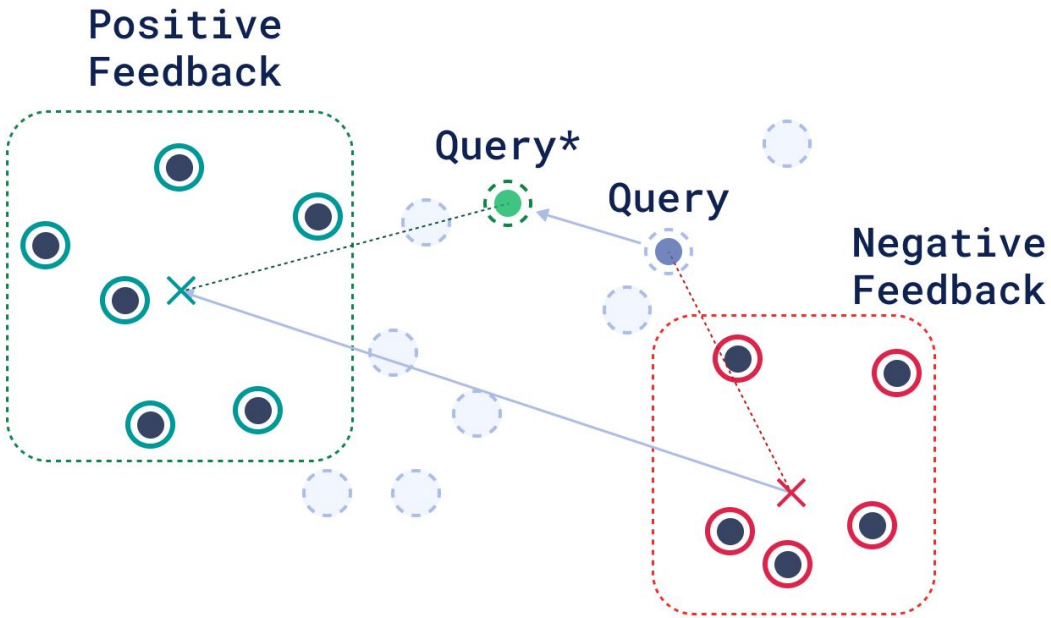
Also can be done on vectors (sparse vectors) but that's a topic for another talk :)

What to Do with Feedback? Change Query



Detailed Article on **Relevance Feedback Methods**

Changing Query as a Vector Example

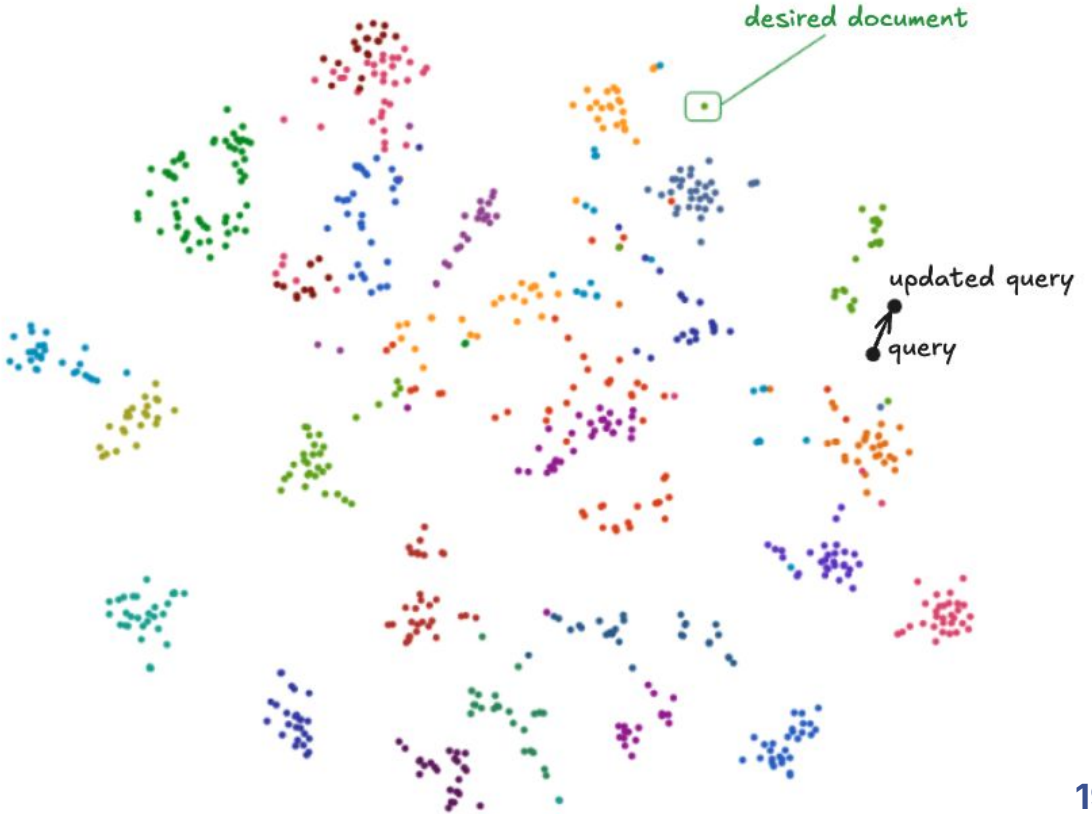


Detailed Article on
Relevance Feedback Methods

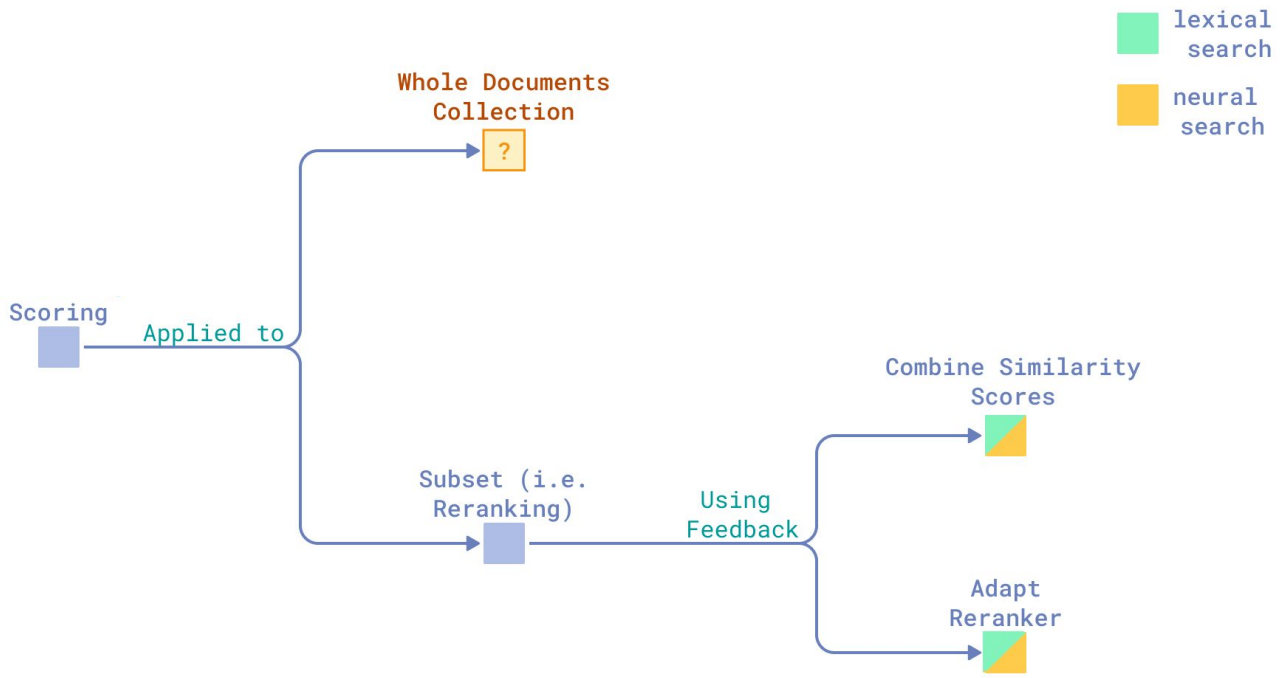
Why Wasn't Query Modification Interesting for Us?

Query is more on the **application side territory**,
less on the **vector search engine** one,

while **search algorithm IS** our concern.



What to do with Feedback? Change Search Algorithm*



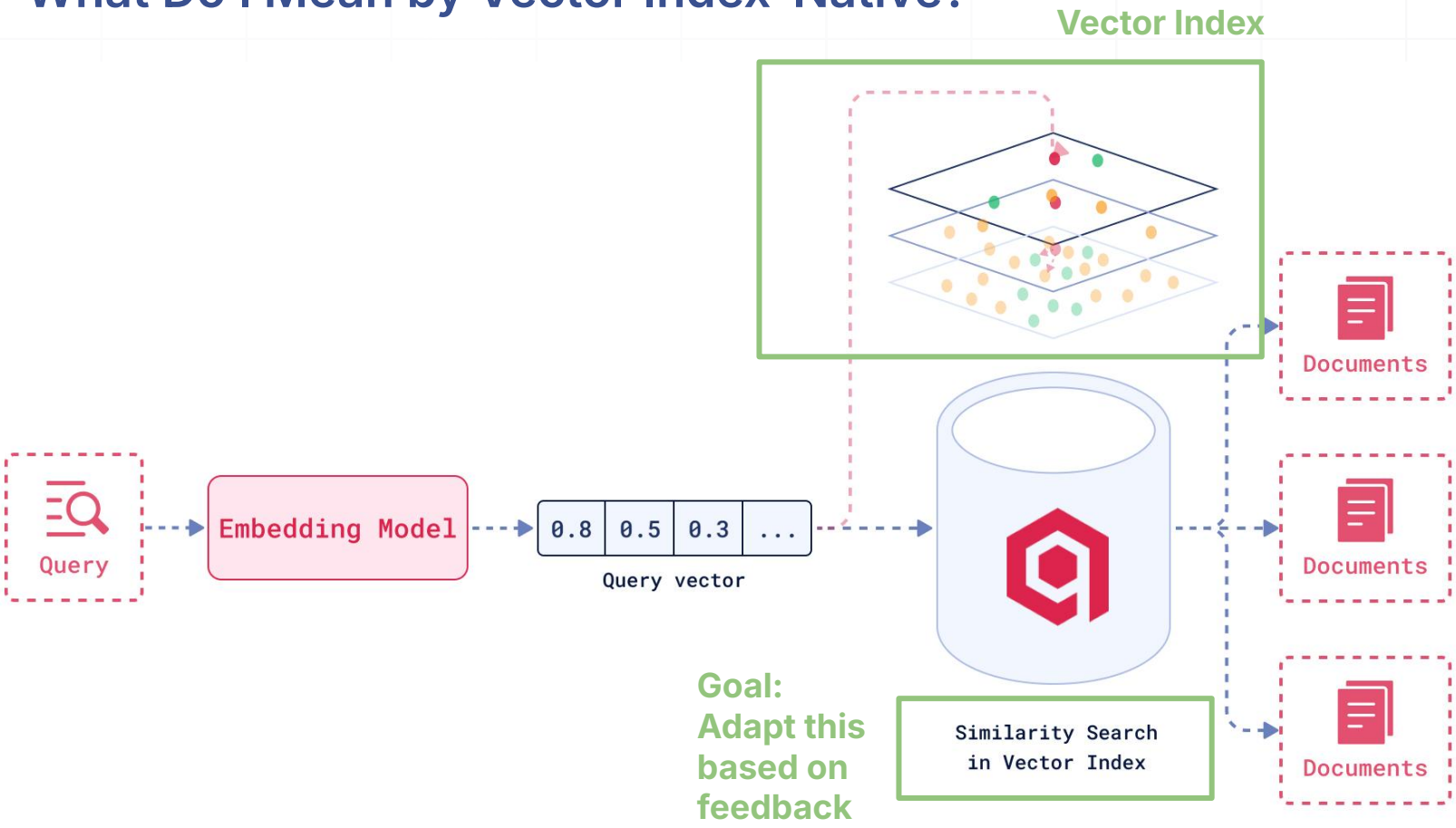
Detailed Article on **Relevance Feedback Methods**

* Meaning function **scoring** how well your request matches your results.

Retrieval surfaces the **data that matches your request the best**, e.g. with the **highest score** (*usually*).

Vector Index-Native Relevance Feedback

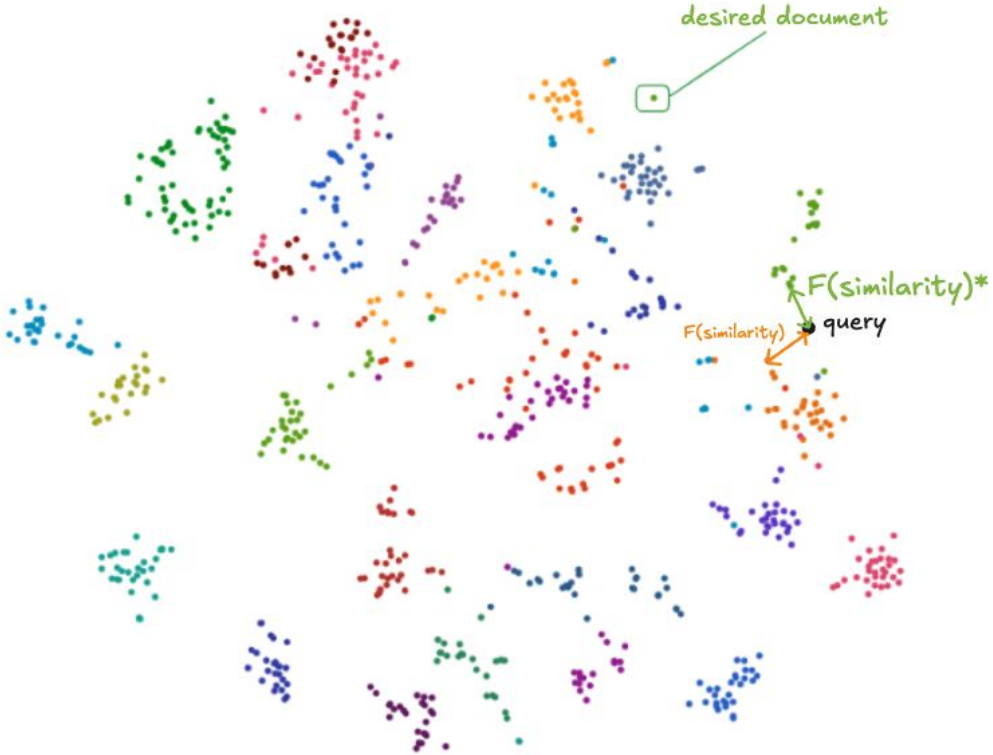
What Do I Mean by Vector Index-Native?



Semantic Similarity Search in Vector Space (Index)

Retrieval = surfacing data with the **best function F score** measured during **search** (e.g. vector index traversal)

Here F = F (similarity) = closeness in vector space, usually measured with **cosine similarity**

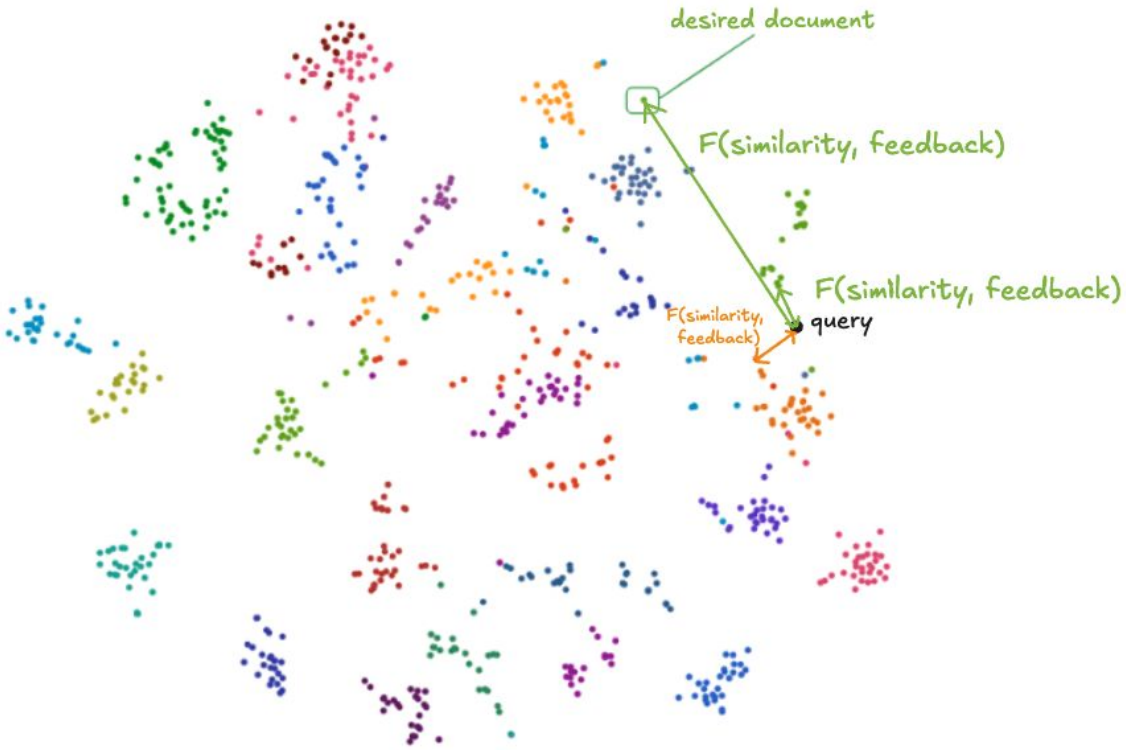


So What to Do with Feedback? Change F(similarity)!

Retrieval = surfacing data with the **best function F score** measured during **search** (e.g. vector index traversal)

What if we incorporated **feedback directly into** the **function F**?

F(similarity) → F(similarity, feedback)



But How...?

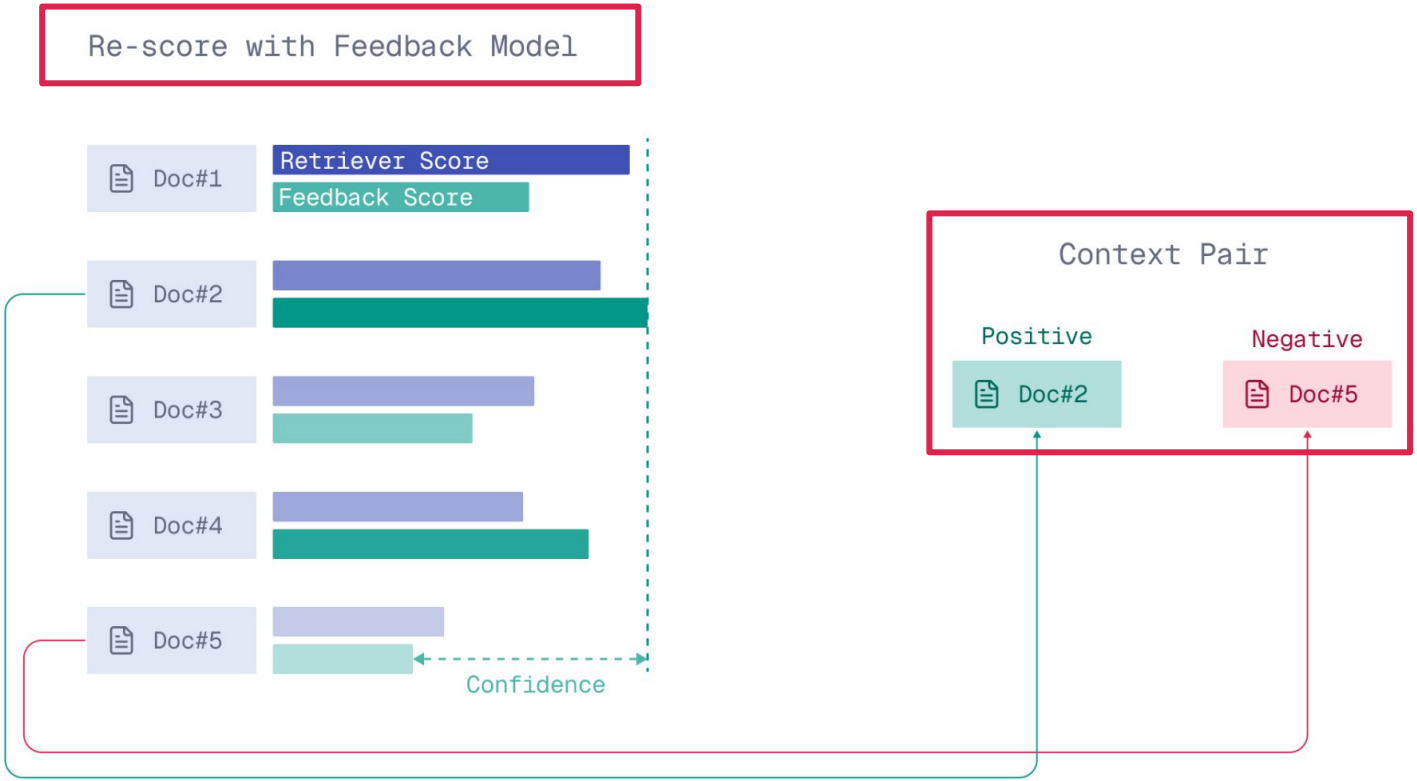
How to extract signals from feedback and adapt **F(similarity) to F(similarity, feedback)** under the hood? And so it's:

- **Cheap**
  - Works with **little feedback**;
  - **No human labeling** required (cause users won't provide it, *trust me*)
- **Universal ⇒** easily **customizable** to every:
  - Dataset;
  - Feedback model;
  - Embedding model (used for retrieval).
- **Working*** in production (**scalable**)
  - Even if retrieved results have **nothing clearly relevant or irrelevant**

∗Working ⇒ you use this & you actually see the improvement, empirically & on metrics

Detailed Article on
Vector Index-Native Relevance Feedback

Introducing Context Pairs



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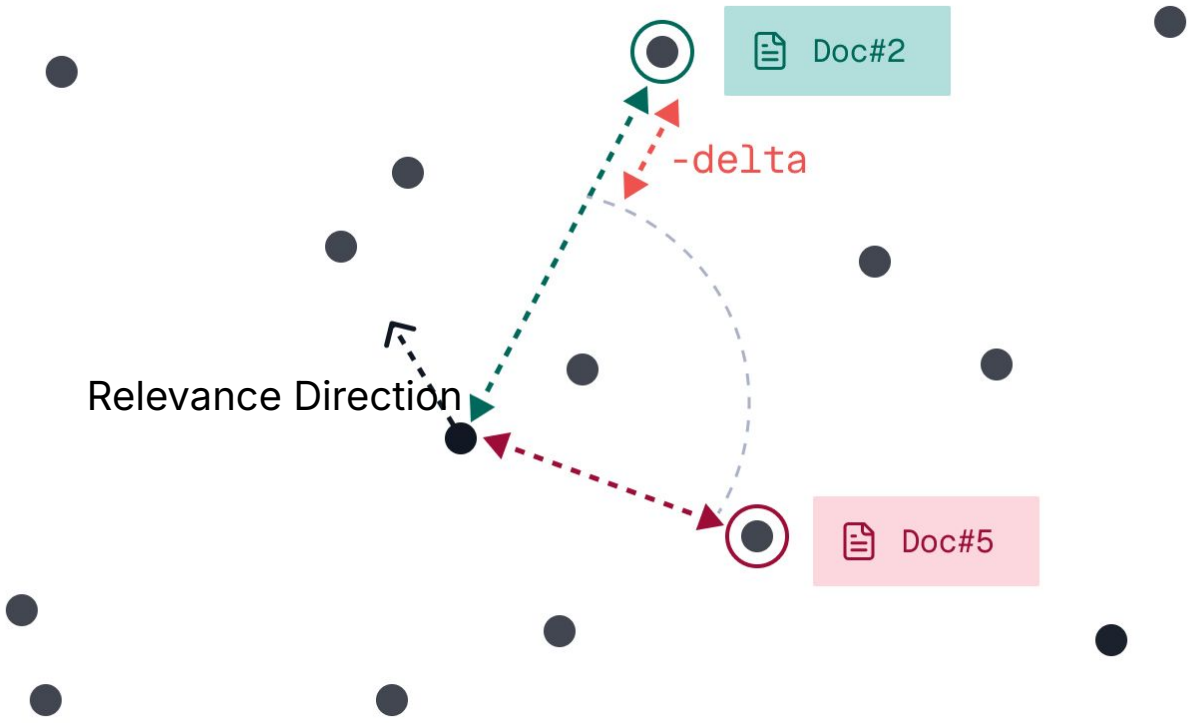
Vector Index-Native Relevance Feedback

Context Pairs Provide Relevance Direction in Vector Space



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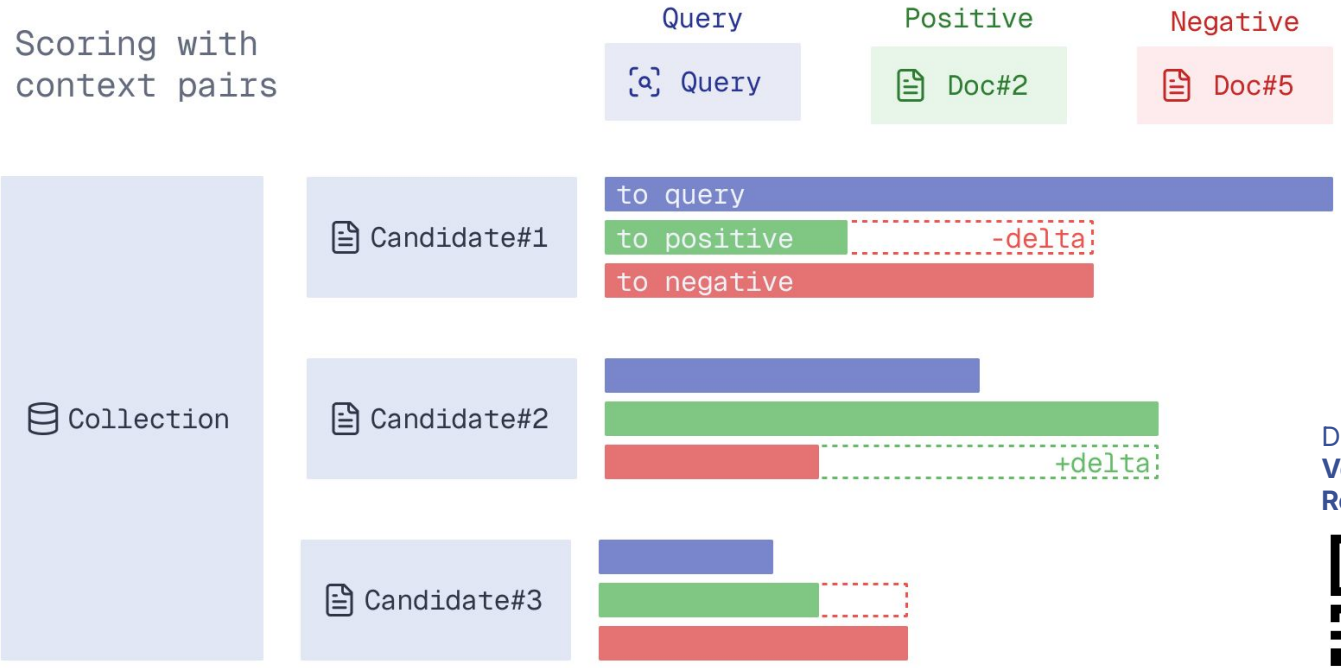
F(similarity, feedback) → F(similarity, context pairs)



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F(similarity, context pairs... & 3 parameters)

Scores will differ per model and dataset.
How to customize? Introduce parameters & train.

F(similarity, context pairs) = **a** * semantic_sim (to document) + avg_sum [confidence (of a context pair) ^ **b** * **c** * delta (document's distance to a context pair)]

a, b, c – just 3 parameters, so **50–100 data points are enough** & **training runs super fast** ⇒easy to adapt

Training objective: Rank as well as a feedback model (**Ranking Loss**)

When you obtain **a, b, c** tuned to your dataset, retriever & feedback model, **plug these a, b, c in each retrieval iteration** (same a, b, c for every query)

Detailed Article on
Vector Index-Native Relevance Feedback

Proving That Vector Index-Native Relevance Feedback Works

Experiments to Prove It

1. Take different datasets, retrievers & feedback models;
2. Do vanilla retrieval without feedback;
3. Do retrieval with relevance feedback formula F(similarity, context pairs & 3 parameters);
4. Show relevance gain.

How to show gain?
Did we surface **more relevant stuff** than the retriever could on its own? Feedback model is the judge.

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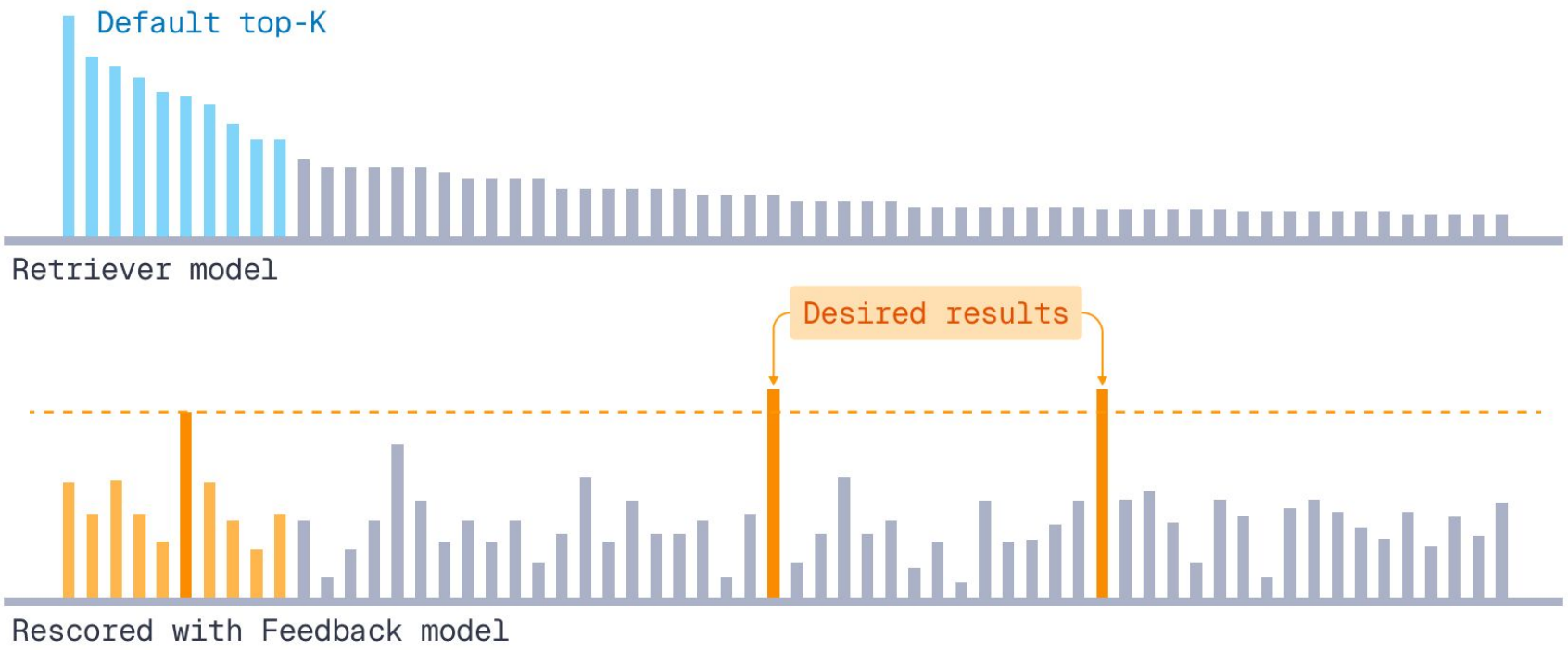
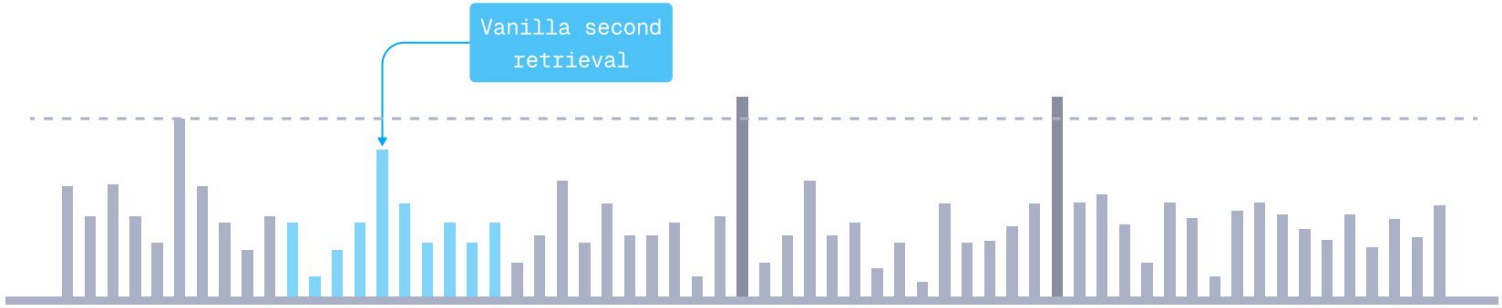
Vector Index-Native Relevance Feedback

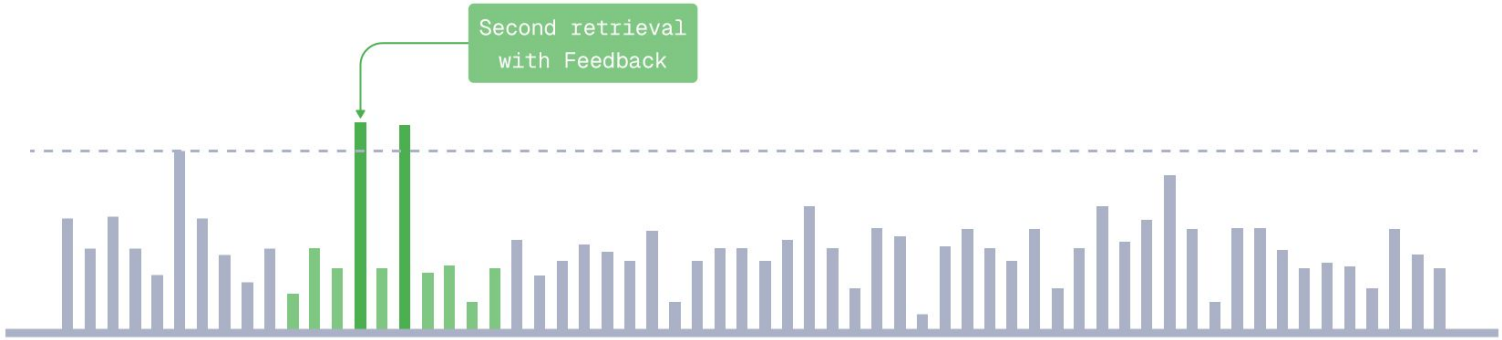
Relevance Gain per Query: Feedback Model is the Judge



Relevance Gain per Query



Retriever model, abovethreshold@10 = 0



Relevance Feedback Formula, abovethreshold@10 = 2

Benchmarks (on a Set of Queries)

	Qwen3-0.6B → colBERTv2.0	Qwen3-0.6B → Qwen3-4B	mxbai-large-v1 → colBERTv2.0
NFCorpus	+10.34%	+10.61%	**+21.57%**
FiQA-2018	+6.45%	+10.94%	**+12.24%**
SCIDOCS	**+38.72%**	+0.69%	+9.55%
MSMARCO	**+23.23%**	+16.73%	+2.40%
Quora	**+5.04%**	+2.67%	0.00%

Using RF to Improve Semantic Search on Documentation

Query: "*recommendations API, how to use*"?

--- Initial Retrieval Results ---

[1] Recommendation API
[2] recommendation API *<chunk duplicate>*.
[3] So, even when the API is not called recommend, recommendation systems can also use this approach and adapt it for their specific use-cases.

--- Additional Results from Relevance Feedback-based Retrieval ---

[1] Recap of the old recommendations API
[2] The result of this API contains one array per recommendation requests.
[3] Deliver Better Recommendations with Qdrant's new API

End-to-End Tutorial

Thank you!

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